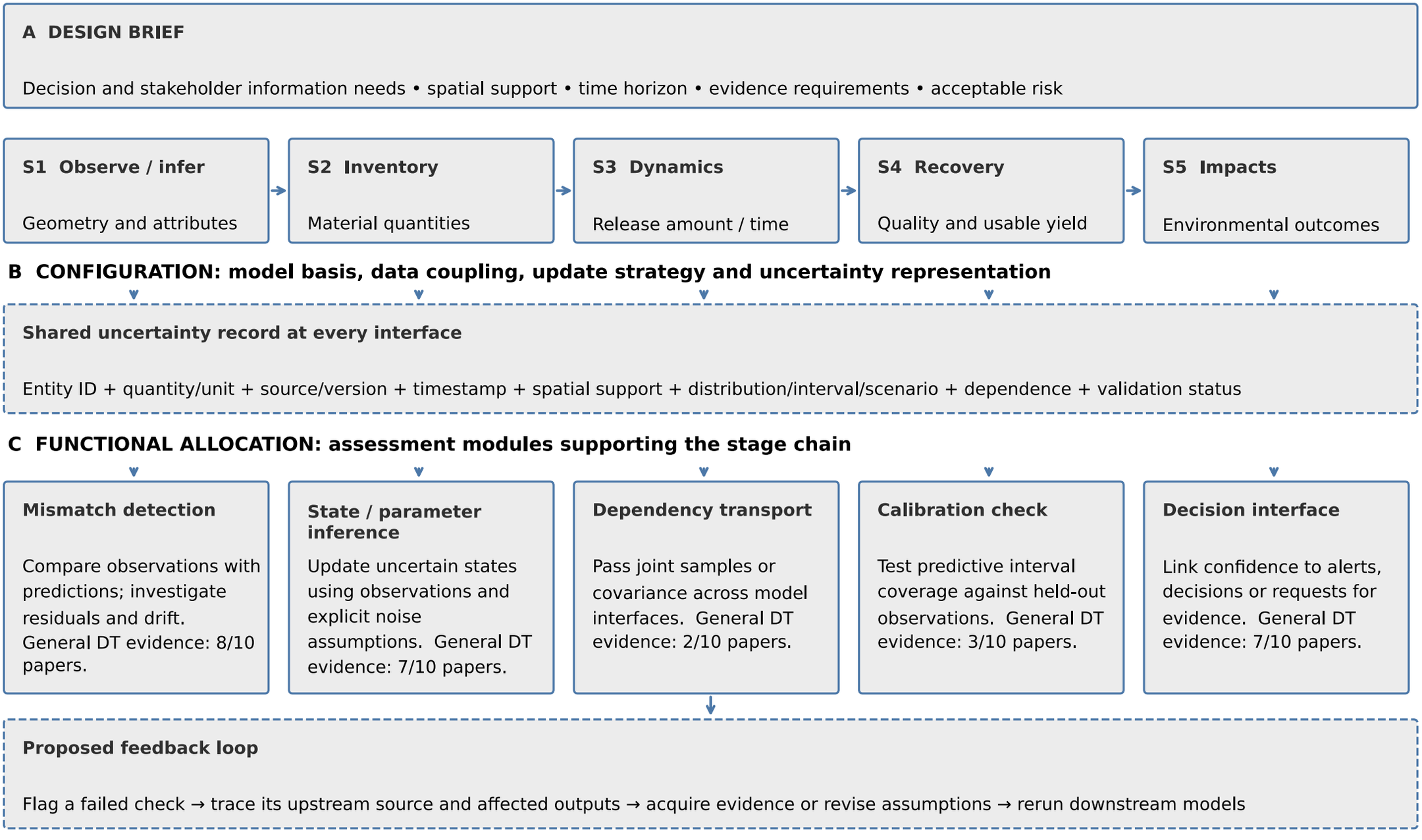
A digital twin architecture for uncertainty continuity

Literature-informed synthesis: supported mechanisms + proposed cross-stage connections



Green: mechanisms coded in general DT studies. Dashed amber: our proposed integration across S1–S5.

Counts are non-exclusive paper frequencies, not accuracy, uncertainty reduction or proof of deployment in the material chain.

Source IDs and coding anchors: architecture_module_evidence.csv. Existing coding reused; no claim of a new independent full-text audit.